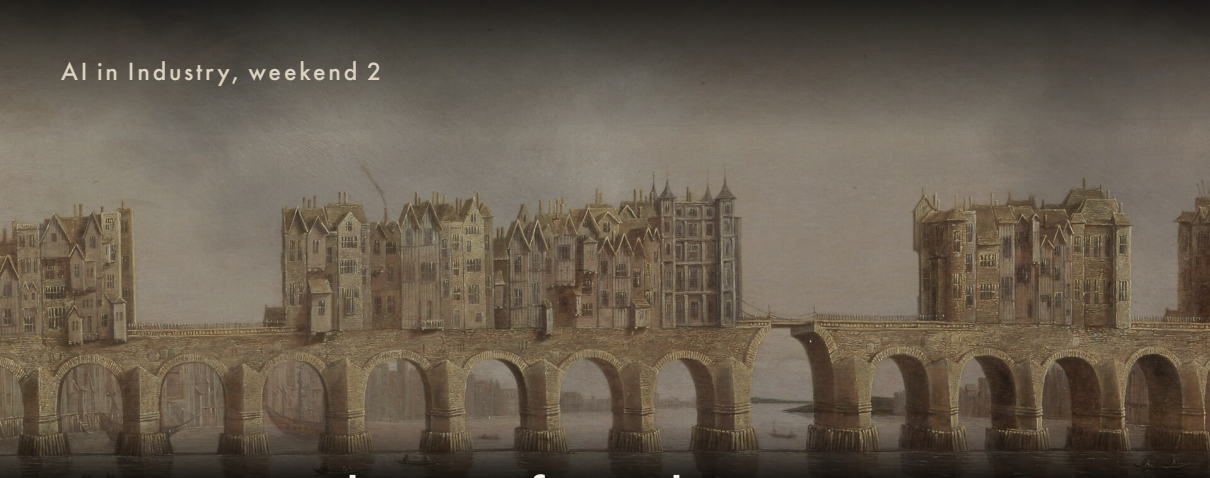
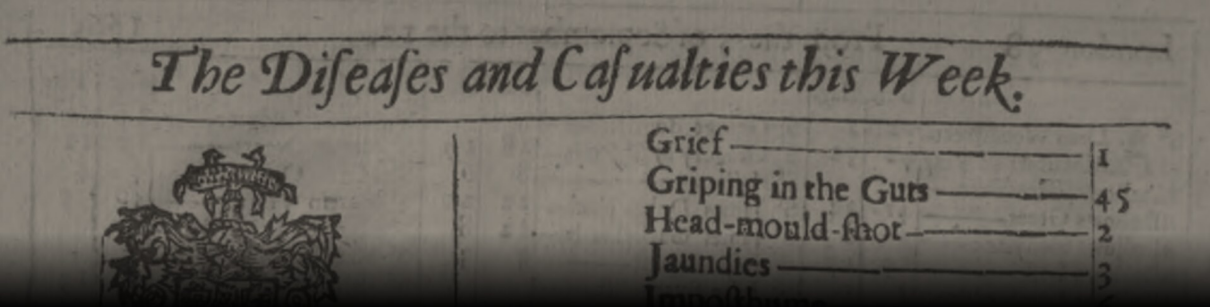
AI in Industry, weekend 2



# Estimating chances from the past

*A book from London, 1662*

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

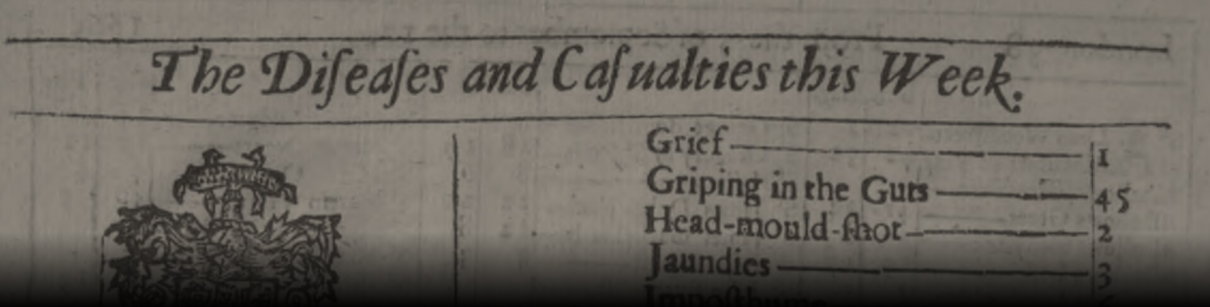



# The Bills of Mortality

London, seventeenth century

Every week, the parish clerks of London print the burials of the week and what each person died of. Londoners read these bills in fear of the plague and of many other diseases.

Parish Clerks of London, bill of 5 to 12 September 1665, detail.
US National Library of Medicine. Internet Archive, public domain



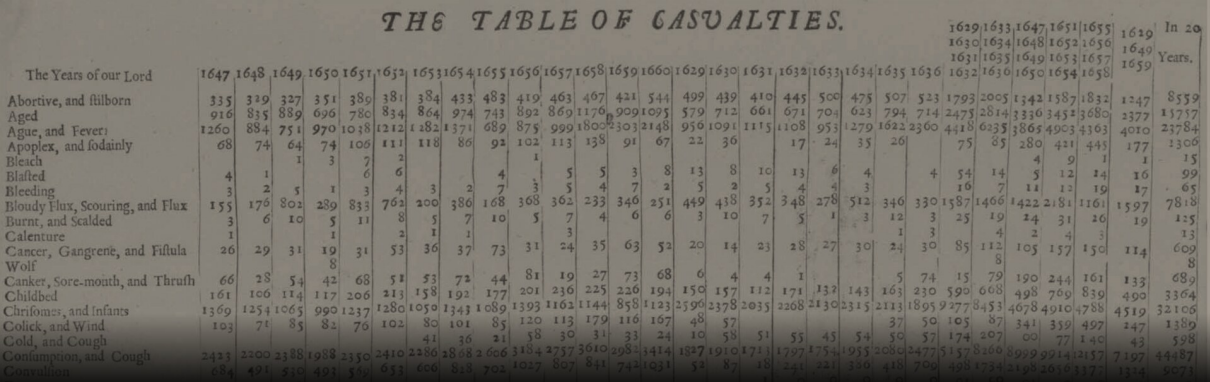

# The Bills of Mortality

London, seventeenth century

Every week, the parish clerks of London print the burials of the week and what each person died of. Londoners read these bills in fear of the plague and of many other diseases.

How afraid should a Londoner be of each disease?

Parish Clerks of London, bill of 5 to 12 September 1665, detail.
US National Library of Medicine. Internet Archive, public domain


THE TABLE OF CASUALTIES.

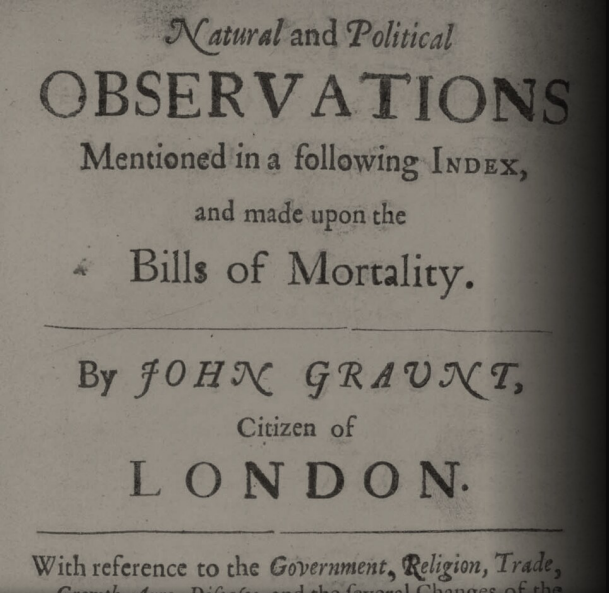
| The Years of our Lord | 1647 | 1648 | 1649 | 1650 | 1651 | 1652 | 1653 | 1654 | 1655 | 1656 | 1657 | 1658 | 1659 | 1660 | 1629 | 1630 | 1631 | 1632 | 1633 | 1634 | 1635 | 1636 | 1629 1630 1631 1632 | 1633 1634 1635 1636 | 1647 1648 1649 1650 | 1651 1652 1653 1654 | 1655 1656 1657 1658 | 1629 1649 1659 | In 20 Years. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Abortive, and ſtilborn | 335 | 329 | 327 | 351 | 389 | 381 | 384 | 433 | 483 | 419 | 463 | 467 | 421 | 544 | 499 | 439 | 410 | 445 | 500 | 475 | 507 | 523 | 1793 | 2005 | 1342 | 1587 | 1832 | 1247 | 8559 |
| Aged | 916 | 835 | 889 | 696 | 780 | 834 | 864 | 974 | 743 | 892 | 869 | 1176 | 909 | 1095 | 579 | 712 | 661 | 671 | 704 | 623 | 794 | 714 | 2475 | 2814 | 3336 | 3452 | 3680 | 2377 | 15757 |
| Ague, and Feveri | 1260 | 884 | 751 | 970 | 1038 | 1212 | 1182 | 1371 | 689 | 875 | 999 | 1800 | 2303 | 2148 | 956 | 1091 | 1115 | 1108 | 953 | 1279 | 1622 | 2360 | 4418 | 6235 | 3865 | 4903 | 4363 | 4010 | 23784 |
| Apoplex, and ſodainly | 68 | 74 | 64 | 74 | 106 | 111 | 118 | 86 | 92 | 102 | 113 | 158 | 91 | 67 | 22 | 36 | | 17 | 24 | 35 | 26 | | 75 | 85 | 280 | 421 | 445 | 177 | 1306 |
| Bleach | | | 1 | 3 | 7 | 2 | | | | 1 | | | | | | | | | | | | | | | 4 | 9 | 1 | 1 | 15 |
| Blaſted | 4 | 1 | | | 6 | 6 | | | 4 | | 5 | 5 | 3 | 8 | 13 | 8 | 10 | 13 | 6 | 4 | | 4 | 54 | 14 | 5 | 12 | 14 | 16 | 99 |
| Bleeding | 3 | 2 | 5 | 1 | 3 | 4 | 3 | 2 | 7 | 3 | 5 | 4 | 7 | 2 | 5 | 2 | 5 | 4 | 4 | 3 | | | 16 | 7 | 11 | 12 | 19 | 17 | 65 |
| Bloudy Flux, Scouring, and Flux | 155 | 176 | 802 | 289 | 833 | 762 | 200 | 386 | 168 | 368 | 362 | 233 | 346 | 251 | 449 | 438 | 352 | 348 | 278 | 512 | 346 | 330 | 1587 | 1466 | 1422 | 2181 | 1161 | 1597 | 7818 |
| Burnt, and Scalded | 3 | 6 | 10 | 5 | 11 | 8 | 5 | 7 | 10 | 5 | 7 | 4 | 6 | 6 | 3 | 10 | 7 | 5 | 1 | 3 | 12 | 3 | 25 | 19 | 24 | 31 | 26 | 19 | 125 |
| Calenture | 1 | | | 1 | | 2 | 1 | 1 | | | 3 | | | | | | | | | | 1 | 3 | | 4 | 2 | 4 | 3 | | 13 |
| Cancer, Gangrene, and Fiſtula | 26 | 29 | 31 | 19 | 31 | 53 | 36 | 37 | 73 | 31 | 24 | 35 | 63 | 52 | 20 | 14 | 23 | 28 | 27 | 30 | 24 | 30 | 85 | 112 | 105 | 157 | 150 | 114 | 609 |
| Wolf | | | | 8 | | | | | | | | | | | | | | | | | | | | 8 | | | | | 8 |
| Canker, Sore-mouth, and Thruſh | 66 | 28 | 54 | 42 | 68 | 51 | 53 | 72 | 44 | 81 | 19 | 27 | 73 | 68 | 6 | 4 | 4 | 1 | | 5 | 74 | 15 | 79 | 190 | 244 | 161 | 133 | 689 |
| Childbed | 161 | 106 | 114 | 117 | 206 | 213 | 158 | 192 | 177 | 201 | 236 | 225 | 226 | 194 | 150 | 157 | 112 | 171 | 132 | 143 | 163 | 230 | 590 | 668 | 498 | 769 | 839 | 490 | 3364 |
| Chriſoms, and Infants | 1369 | 1254 | 1065 | 990 | 1237 | 1280 | 1050 | 1343 | 1089 | 1393 | 1162 | 1144 | 858 | 1123 | 2596 | 2378 | 2035 | 2208 | 2130 | 2315 | 2113 | 1895 | 9277 | 8453 | 4078 | 4910 | 4788 | 4519 | 32106 |
| Colick, and Wind | 103 | 71 | 85 | 82 | 76 | 102 | 80 | 101 | 85 | 120 | 113 | 179 | 116 | 167 | 48 | 57 | | | | | 37 | 50 | 105 | 87 | 341 | 359 | 497 | 247 | 1389 |
| Cold, and Cough | | | | | | 41 | 36 | 21 | 58 | 30 | 31 | 33 | 24 | 10 | 58 | 51 | 55 | 45 | 54 | 50 | 57 | 174 | 207 | 60 | 77 | 140 | 43 | 598 |
| Conſumption, and Cough | 2423 | 2200 | 2388 | 1988 | 2350 | 2410 | 2280 | 2808 | 2666 | 3184 | 2757 | 3610 | 2982 | 3414 | 1827 | 1910 | 1713 | 1797 | 1754 | 1955 | 2080 | 2477 | 5157 | 8200 | 8999 | 9914 | 12137 | 7197 | 44487 |
| Convulſion | 684 | 491 | 510 | 493 | 569 | 653 | 600 | 828 | 702 | 1027 | 807 | 841 | 742 | 1031 | 51 | 87 | 18 | 241 | 221 | 330 | 418 | 709 | 498 | 1734 | 2198 | 2656 | 3377 | 1314 | 9071 |

# I shall onely set down how many died of each.

*John Graunt, a haberdasher of London, 1662*

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2

John Graunt, *Natural and Political Observations*, 1662, the Table of Casualties, detail. Wikimedia Commons, public domain


*Natural* and *Political*

OBSERVATIONS

Mentioned in a following INDEX,

and made upon the

Bills of Mortality.

By JOHN GRAUNT,

Citizen of

LONDON.

With reference to the *Government*, *Religion*, *Trade*, *Growth*, *Ayre*, *Diseases*, and the several Changes of the

John Graunt, *Natural and Political Observations*, 1662, title page, detail. Wikimedia Commons, public domain

# John Graunt

1620 to 1674, haberdasher

"Now having (I know not by what accident) engaged my thoughts upon the Bills of Mortality, and so far succeeded therein, as to have reduced several great confused Volumes into a few perspicuous Tables…"

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, the dedication

# Of 229250 deaths

twenty years of bills

| | |
|---|---|
| Drowned | 829 |
| Sodainly | 454 |
| Grief | 279 |
| Lunatique | 158 |
| Murthered | 86 |
| Head-Ach | 51 |
| Frighted | 22 |
| Excessive drinking | 2 |

"That not one in two thousand are Murthered in London."

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2 and the index

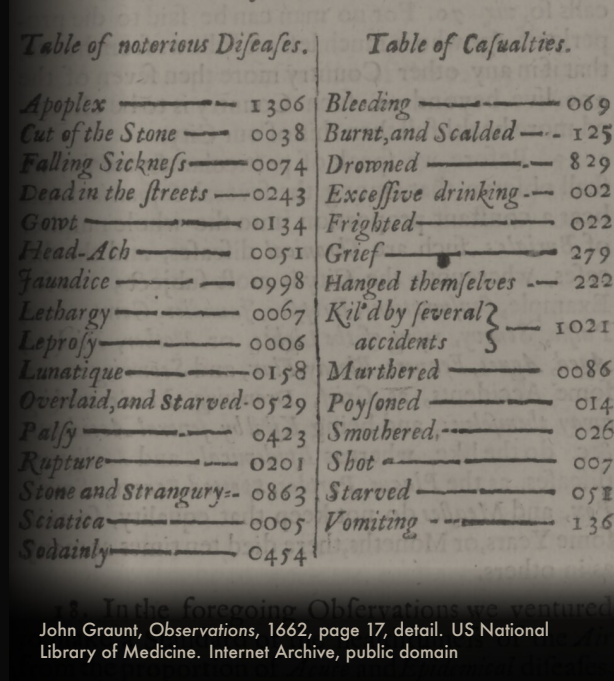


*Table of noterious Diseases.*

| | |
|---|---|
| Apoplex | 1306 |
| Cut of the Stone | 0038 |
| Falling Sickness | 0074 |
| Dead in the streets | 0243 |
| Gowt | 0134 |
| Head-Ach | 0051 |
| Jaundice | 0998 |
| Lethargy | 0067 |
| Leprosy | 0006 |
| Lunatique | 0158 |
| Overlaid, and Starved | 0529 |
| Palsy | 0423 |
| Rupture | 0201 |
| Stone and Strangury | 0863 |
| Sciatica | 0005 |
| Sodainly | 0454 |

*Table of Casualties.*

| | |
|---|---|
| Bleeding | 069 |
| Burnt, and Scalded | 125 |
| Drowned | 829 |
| Excessive drinking | 002 |
| Frighted | 022 |
| Grief | 279 |
| Hanged themselves | 222 |
| Kil'd by several accidents | 1021 |
| Murthered | 0086 |
| Poysoned | 014 |
| Smothered | 026 |
| Shot | 007 |
| Starved | 051 |
| Vomiting | 136 |

13. In the foregoing Observations we ventured

John Graunt, *Observations*, 1662, page 17, detail. US National Library of Medicine. Internet Archive, public domain



"Whereas many persons live in great fear, and apprehension of some of the more formidable, and notorious diseases... I shall onely set down how many died of each: that the respective numbers, being compared with the Total 229250, those persons may the better understand the hazard they are in."

John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain



"Whereas many persons live in great fear, and apprehension of some of the more formidable, and notorious diseases... I shall onely set down how many died of each: that the respective numbers, being compared with the Total 229250, those persons may the better understand the hazard they are in."

Graunt divides each count by all the deaths.
From the past, every Londoner can estimate the hazard of each disease.

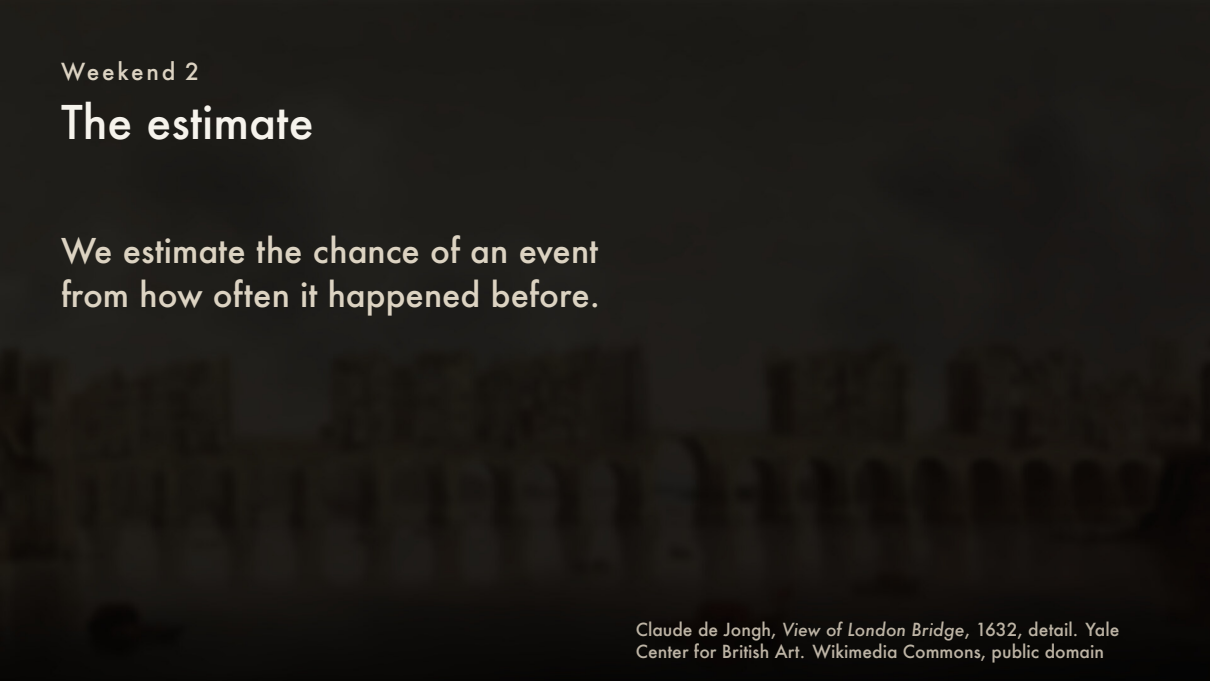
John Graunt, *Natural and Political Observations... upon the Bills of Mortality*, London, 1662, chapter 2

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

# The estimate

We estimate the chance of an event from how often it happened before.



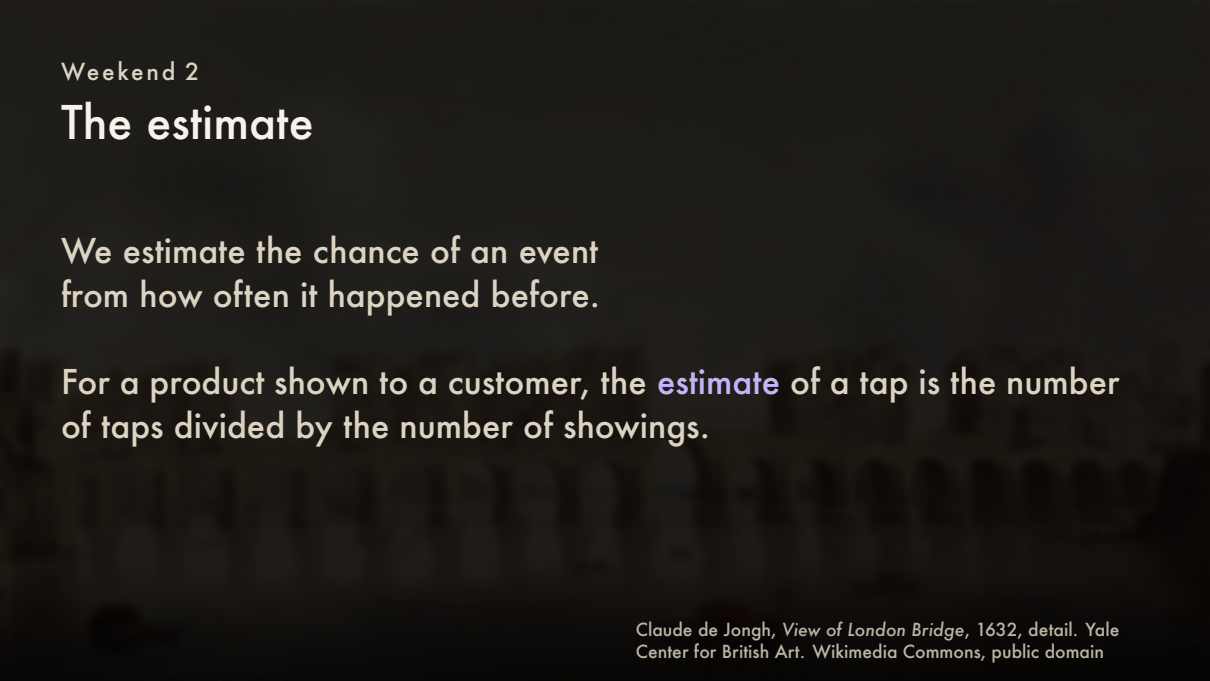
Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

# The estimate

We estimate the chance of an event from how often it happened before.

For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.



Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

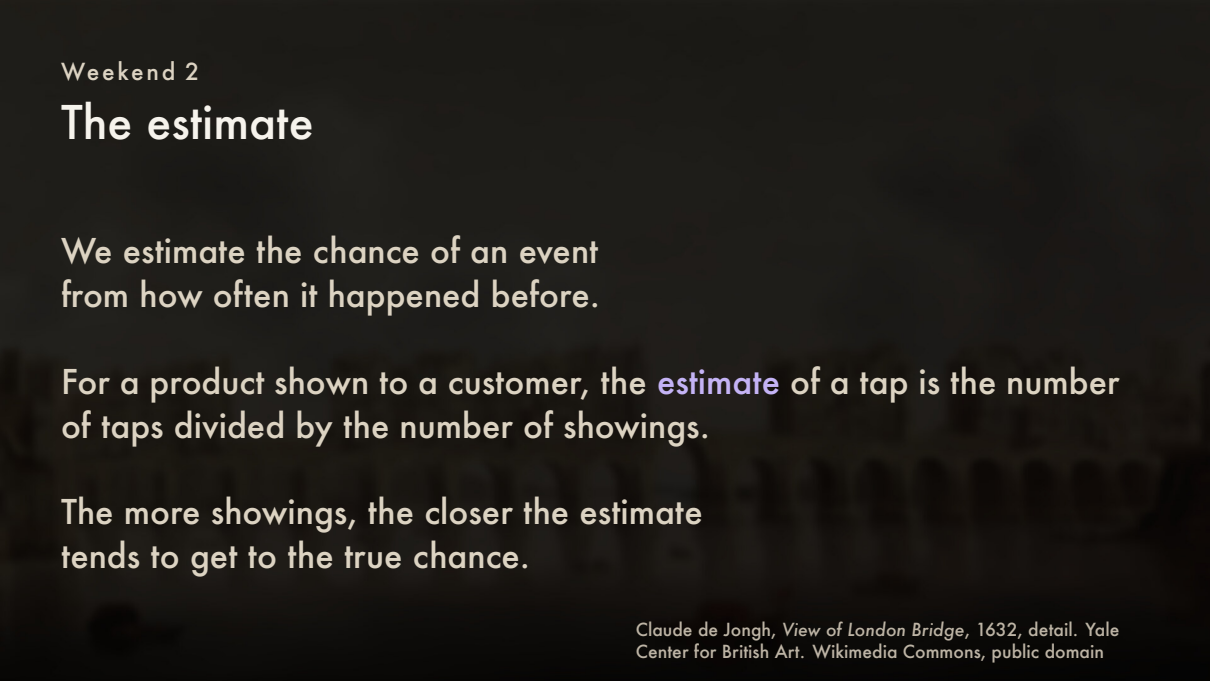
Weekend 2

# The estimate

We estimate the chance of an event
from how often it happened before.

For a product shown to a customer, the estimate of a tap is the number of taps divided by the number of showings.

The more showings, the closer the estimate
tends to get to the true chance.



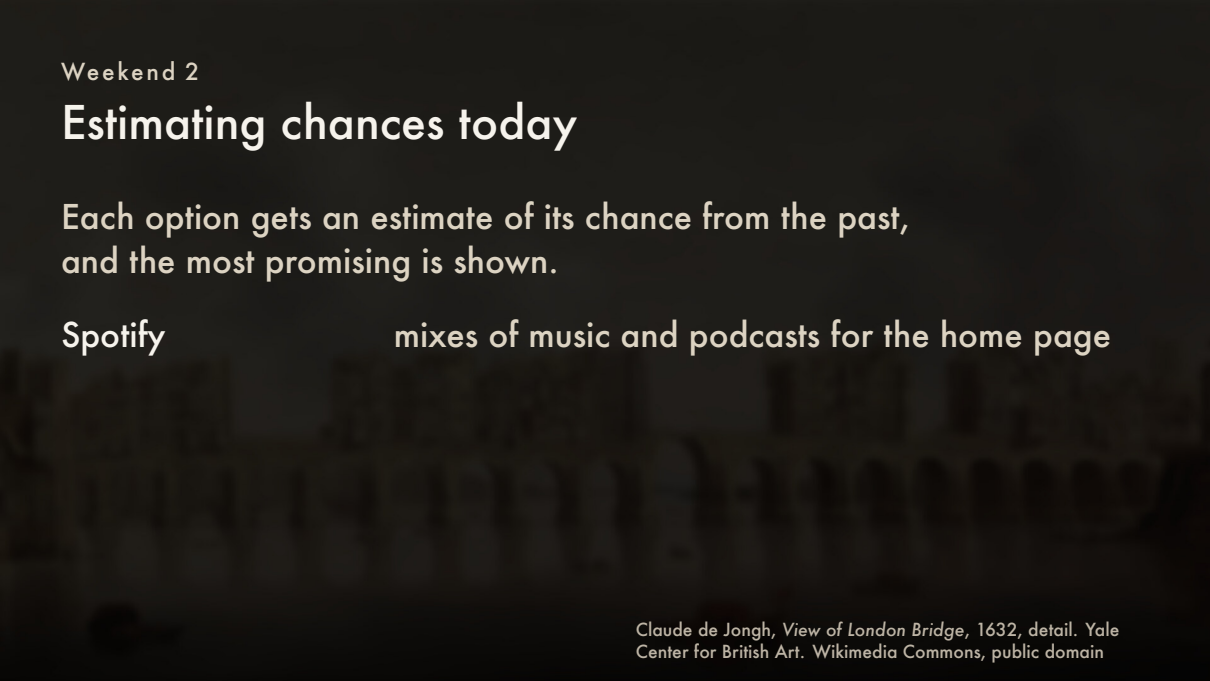
Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

# Estimating chances today

Each option gets an estimate of its chance from the past,
and the most promising is shown.

Spotify mixes of music and podcasts for the home page



Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

# Estimating chances today

Each option gets an estimate of its chance from the past,
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| | |
|---|---|
| Spotify | mixes of music and podcasts for the home page |
| Amazon Payments | images and taglines for a page |

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

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Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

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Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain

Weekend 2

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| Online shops | products for a customer |
| News sites | headlines for a story |

Such estimates decide much of what we are shown online.

Claude de Jongh, *View of London Bridge*, 1632, detail. Yale Center for British Art. Wikimedia Commons, public domain